

The Ministry of Higher Education and Scientific Research		Sub.: ECE304 Seq. Logic Circuit Year: Third
The University of Kufa College of Engineering		Date: Nov 22, 2023 Exam Duration: 2 Hours
Department of Electronic and Communications Engineering		Examiner: Wahhab R. Mousa

Fall Semester 2023-2024, Midterm Exam.

Notes: (1) Try FIVE questions; (2) Use ONLY English; and (3) Calculator is NOT allowed.



Figure 1: T_0



Figure 2: CLK

Q1: (20 Marks)

Using four JK-type flip-flops and any number of multiplexers, design a 4-bit counter-register with mode selection as follows:

s_1	s_0	Register Operation
0	0	No change
1	0	Counts up
0	1	XOR the input with current state of the flip flops
1	1	Load parallel data

Q2: (20Marks)

It is necessary to generate a repeated timing signal like T_0 shown in Figure 1 above. The frequency of the CLK is 80MHz. The time duration between pulses of T_0 is six times the duration of one cycle of CLK. Design the circuit using: (a) flip-flops only. (b) a counter and a decoder.

Q3: (20Marks)

Design a counter with T flip-flops that goes through the following binary repeated sequence: 0, 1, 3, 7, 6, 4. Show that when binary states 010 and 101 are considered as don't care conditions, the counter may not operate properly. Find a way to correct the design.

Q4: (20Marks)

A 16K * 4 memory uses coincident decoding by splitting the internal decoder into X-selection and Y-selection. (a) What is the size of each decoder, and how many AND gates are required for decoding the address? (b) Determine the X and Y selection lines that are enabled when the input address is the binary equivalent of 91.

Q5: (20 Marks)

(a) How many 16K * 8 RAM chips are needed to provide a memory capacity of 128K * 16? Draw the required memory and label all lines clearly. (b) How many lines of the address must be used to access 128K * 16? How many of these lines are connected to the address inputs of all chips? (c) How many lines must be decoded for the chip select inputs? Specify the size of the decoder.

Q6: (20 Marks)

Show that the best practice for coincident decoding is to divide the address lines by two. Otherwise, the coincident decoding technique doesn't achieve an optimum practice.